Dynamic Event-Triggered Consensus Control for Leader-Following Multiagent Systems With Communication Delay

Xiaonan Lin, Fang Wang, Chao Zhou, and Wen Liu

***Abstract*—This brief considers the dynamic event-triggered consensus control design of leader-following multiagent systems (MASs) under communication delay. Static event-triggered mechanism is often used, but its energy saving is limited. The dynamic event-triggered mechanism based on sampled data can obtain fewer trigger times than static event-triggered mechanism by adding a dynamic variable, which means that energy is saved to a greater extent and Zeno behaviors can be excluded obviously. Different from the existing works, this brief considers time-varying communication delay between agents. A periodic switching controller is used to deal with delay. Besides, the Lyapunov-Krasovskii function is applied to prove the stability of the closed-loop system and linear matrix inequality plays an important role in this process. Finally, simulation results illustrate the availability of the proposed controller.**

***Index Terms*—Dynamic event-triggered, consensus, multiagent systems, time-varying communication delay, linear matrix inequality.**

I. Introduction

IN RECENT years, MASs have been applied in many fields, such as unmanned aerial vehicles [1], multi robots [2] and smart-homes controlled by heuristic algorithms [3]–[5], which are not affected by the minimum. All of them have brought convenience to our life. So in the field of control, we often consider how to optimize the algorithm [6]–[8], and whether the system has disturbances [9] or constraints [10], but it is important to consider the consistency [11]–[12] because agents need to communicate. To avoid continuous communication, the event-triggered mechanism (ETM) came into being.

Generally speaking, ETM can be divided into static event-triggered mechanism (SEM) [13]–[14] and dynamic event-triggered mechanism (DEM) [15]–[16]. SEM has been widely used in MASs, but some packets may be lost during transmission. Based on SEM, a DEM, which involves internal dynamic variables, has been studied. In [17], a DEM was proposed, which the threshold parameter could be adjusted dynamically according to the dynamic law, compared with SEM, the system could reduce the number of triggers on the premise of achieving consensus. A DEM was proposed in [18] to decrease the communication between agents, and then an observer was designed by sampling the agent state, the effect of delay was eliminated. A distributed adaptive event-triggered fault tolerant control of MASs was studied in [19]. In the above results, the decay rate of the dynamic variables is relatively fast, but the DEM in this brief can update the controller less times by a dynamic variable with slow attenuation. Besides, the network congestion is reduced and the cost is saved.

Although the results in [17]–[19] reduce the cost, due to the consistency of MASs need to be achieved through local computing and communication, it is necessary to study the communication delay when considering the consistency. In [20], the time-varying delay between MASs was considered and adopted a periodic controller to deal with this delay. In [21], the consistency of MASs with constant delay was considered. The cluster formation control of MASs with time-varying communication delay was considered in [22], but it weakened the influence of some delay on the control system. This brief investigates the reasonable and complex time-varying communication delay. The delay will cause asynchronous phenomenon of signal transmission between MASs, a switching control method is designed to avoid the problem.

Inspired by the above, the main contributions are as follows: (1) Compared with [21]–[22], we consider a set of MASs with a single leader and a nonuniform time-varying communication delay. It has a wider applicability, and a switching controller which divides the time into time delay and non time delay, and considers the stability respectively; (2) Compared with [13]–[14], a DEM which can effectively reduce the trigger times by slowing down the decay rate of the dynamic variables is considered; (3) Most of the literatures on linear MASs don't consider the simulation of actual system, as shown in [23], but in this brief, a practical example is given and verified the controller. So the innovation of this brief mainly focuses on eliminating the influence of time-varying communication delay on the system through a switching controller and making the controller get less trigger times via a more effective DEM. The theoretical, practical and the comparative simulation verify the effectiveness and advantages of the controller in Section V.

The structure of this brief is as follows. Section II introduces some basic knowledge. Section III designs a dynamic event-triggered controller. Section IV presents main results. Section V shows simulation result. Section VI gives the summary and prospect of this brief.

II. System Description

A. Notation and Graph Theory

Undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ is considered in this brief, where $\mathcal{V}$ is a collection with nodes $v_1, \ldots, v_N$ as its elements,

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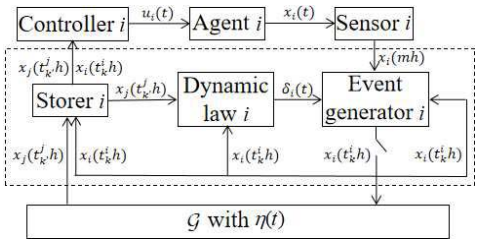


Fig. 1. Work flow of the control scheme.

$\mathcal{E}$ is the collection of edges, if there is a channel connection between node v_i and v_j, then $(v_i, v_j) \in \mathcal{E}$. $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n\times n}$ is the adjacency matrix with $a_{ii} = 0$, $a_{ij} = a_{ji} = 1$ if $(v_i, v_j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. The Laplacian matrix $\mathcal{L}$ is defined as $l_{ii} = \sum_{j=1}^{N} a_{ij}$ and $l_{ij} = -a_{ij}$, $i \neq j$. $\mathcal{M} = diag\{m_1, \ldots, m_N\} \in \mathbb{R}^{n\times n}$, $m_i = 0$ if the ith agent can't receive the signal from leader and $m_i = 1$ otherwise. Define $\mathcal{N} = \mathcal{L} + \mathcal{M}$, its eigenvalues are satisfied $0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$. I denotes an identity matrix and $I = diag\{1, 1, \ldots, 1\} \in \mathbb{R}^{n\times n}$.

B. Multiagent Systems

The dynamics of $N + 1$ linear agents are described in the following form:

$$\begin{cases} \dot{x}_0(t) = Ax_0(t), \\ \dot{x}_i(t) = Ax_i(t) + Bu_i(t), \end{cases} \tag{1}$$

where $x_0(t) \in \mathbb{R}^n$, $x_i(t) \in \mathbb{R}^n$, $u_i(t) \in \mathbb{R}^n$ are the state of the leader, the state and control input of the ith follower, respectively. A and B are constant matrices.

Assumption 1: The pair (A, B) is stabilizable.

Assumption 2: Topology graph $\mathcal{G}$ contains a directed spanning tree with a leader as the root node, and the follower's subgraph is undirected.

This brief designs a dynamic event-triggered controller to make the system achieve exponential stability and there is no Zeno behaviors.

C. Basic Lemmas

Lemma 1 [20]: For positive scalers α, γ, define η as the delay, $\Theta_D(0, t)$ as the time period of all delays and $\Theta_H(0, t) = [0, t] - \Theta_D(0, t)$. If there are $\dot{V}(\varepsilon(t)) < \gamma V(\varepsilon(t))$ when $t \in [mh, mh + \eta]$ and $\dot{V}(\varepsilon(t)) < -\alpha V(\varepsilon(t))$ when $t \in [mh + \eta, (m + 1)h](m \in \mathbb{N})$, then we have $V(\varepsilon(t)) \leq e^{-\alpha|\Theta_H(0,t)|} e^{\gamma|\Theta_D(0,t)|} V(\varepsilon(0))$.

III. Dynamic Event-Triggered Control System Design

A. Dynamic Event-Triggered Mechanism

Fig. 1 is the flow chart of dynamic event-triggered controller. $x_i(mh)$ is the sampling data of agent i with h as the period, $x_j(t_{k'}^j h)$ is transmitted to the storer i through the network $\mathcal{G}$. $x_i(mh)$, $\delta_i(t)$, $x_i(t_k^i h)$ and $x_j(t_{k'}^j h)$ are transmitted to the event generator i to determine whether $x_i(mh)$ will be triggered. When the error $e_i(t) = x_i(mh) - x_i(t_k^i h)$ meets the triggering conditions, $x_i(mh)$ will be transmitted as $x_i(t_k^i h)$ to its neighbor agent. Based on the above, the triggered signals will be transmitted to the controller i, and then the excitation $u_i(t)$ will be generated and transmitted to the agent i, so that the MASs can circulate. Next, the sampled data based on DEM will be given. For $t \in [mh, mh + h)$,

$$\begin{aligned} t_{k+1}^i h = \inf_{m\in N} \{ & mh > t_k^i h | e_i^{\mathrm{T}}(t) \Omega e_i(t) \\ & \geq \theta \zeta_i^{\mathrm{T}}(t_k^i h) \Omega \zeta_i(t_k^i h) + \mu_i \delta_i(t)\}, \end{aligned} \tag{2}$$

$$\dot{\delta}_i(t) = \begin{cases} -\delta_i(t) - \mu_i \delta_i(mh) + \zeta_i^{\mathrm{T}}(t_k^i h) Z \zeta_i(t_k^i h), & \delta_i(t) < \bar{\delta}_i, \\ -\delta_i(t) - \mu_i \delta_i(mh), & \delta_i(t) \geq \bar{\delta}_i, \end{cases} \tag{3}$$

where h is the sampling interval, $\mu_i \in (0, 1)$ and $\theta > 0$,

$$\zeta_i(t_k^i h) = \sum_{j=1}^{N} (a_{ij}(x_i(t_k^i h) - x_j(t_{k'}^j h)) + m_i(x_i(t_k^i h) - x_0(mh))).$$

The designed DEM is based on $e_i(t)$, and a dynamic variable $\delta_i(t)$ is introduced into it, so as to reduce the attenuation speed of $\delta_i(t)$ as much as possible to reduce the number of triggeres on the premise of ensuring the stability of the system. Besides, $\bar{\delta}_i > 0$, $\Omega > 0$ and $Z > 0$ will be designed later.

Lemma 2 [24]: $\mu_i > 0$, $\bar{\delta}_i > 0$ are parameters, $\delta_i(t) \geq 0$ holds if μ_i satisfies $h \leq -\ln \frac{\mu_i}{1+\mu_i}$.

When MASs transform their signals from each other, there is an effect of communication delay. When DEM is satisfied, $x_i(t_k^i h)$ will arrive at its neighbor agent j at $t_k^i h + \eta_k^{ij} + \bar{\eta}$, where $\eta_k^{ij} \leq \bar{\eta}$ is the delay of the channel from agent i to its neighbor agent j and $\bar{\eta}$ is the minimum upper bound of communication delay.

Remark 1: DEM can reduce the trigger better than SEM by adding a dynamic variable. The SEM [25] is a special form of (2) ($\delta_i(t) = 0$). The first equation of (2) is similar to the DEM of [26], but the form of dynamic variable in this brief doesn't make it decay too fast, it can work longer. Moreover, the signal in the DEM needs to be sampled periodically, which excludes Zeno behaviors.

B. Controller Design

The dynamic event-triggered controller is designed as:

$$u_i(t) = \begin{cases} 0, & \eta(t) \in [0, \eta), \\ -K \sum_{j=1}^{N} (a_{ij}(x_i(t_k^i h) - x_j(t_{k'}^j h)) & \\ \quad + m_i(x_i(t_k^i h) - x_0(mh))), & \eta(t) \in [\eta, h), \end{cases} \tag{4}$$

where $\eta(t) = t - mh$ and $\eta = \eta_k^{ij} + \bar{\eta}$, $t \in [mh, mh + h)$. K is the gain matrix and it will be designed later.

Denote $\varepsilon_i(t) = x_i(t) - x_0(t)$, taking the time derivative of $\varepsilon_i(t)$, and substituting (1) and (4) into it, then we can get the closed-loop system of agent i, $(i = 1, 2, \ldots, N)$, for $t \in [mh, mh + h)$ is

$$\dot{\varepsilon}_i(t) = \begin{cases} A\varepsilon_i(t), & \eta(t) \in [0, \eta), \\ A\varepsilon_i(t) - BK \sum_{j=1}^{N} (a_{ij}(x_i(t_k^i h) - x_j(t_{k'}^j h)) & \\ \quad + m_i(x_i(t_k^i h) - x_0(mh))), & \eta(t) \in [\eta, h). \end{cases} \tag{5}$$

Because of $x_i(t_k^i h) - x_j(t_k^j h) = x_i(mh) - e_i(mh) - x_j(mh) + e_j(mh)$, $x_i(t_k^i h) - x_0(mh) = x_i(mh) - e_i(mh) - x_0(mh)$, combining with (5), we get the equation

$$\dot{\varepsilon}(t) = \begin{cases} (I \otimes A)\varepsilon(t), & \eta(t) \in [0, \eta), \\ (I \otimes A)\varepsilon(t) - (\mathrm{N} \otimes BK)\varepsilon(t - \eta(t)) & \\ \quad + (\mathrm{N} \otimes BK)e(mh), & \eta(t) \in [\eta, h), \end{cases}$$

where $\varepsilon(t) = col\{\varepsilon_1(t), \ldots, \varepsilon_N(t)\}$.

IV. Stability Analysis

Theorem 1: For positive scalers $\alpha, \beta, \iota, \theta$ and a controller gain K, if there are matrices $P > 0$, $Q > 0$, $R > 0$, $W > 0$, $\Omega > 0$, $Z > 0$, Υ_1, Υ_2 and μ_i satisfies *Lemma 2* such that

$$\Pi^a(\lambda_i) = \begin{bmatrix} \Pi^a{}_{11} & \Pi^a{}_{12}(\lambda_i) & \Pi^a{}_{13} & \lambda_i PBK & \beta PA \\ * & \Pi^a{}_{22}(\lambda_i) & \Pi^a{}_{23} & \Pi_{24}(\lambda_i) & -\beta\lambda_i PBK \\ * & * & \Pi^a{}_{33} & 0 & 0 \\ * & * & * & \Pi_{44}(\lambda_i) & \beta\lambda_i PBK \\ * & * & * & * & h^2 R - 2\beta P \end{bmatrix} < 0,$$

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$$\Xi^a = \begin{bmatrix} \Xi^a{}_{11} & \Xi^a{}_{12} & e^{-\alpha h}\mathcal{R}^a{}_{21} & 0 & \iota PA \\ * & \Xi^a{}_{22} & \Xi^a{}_{23} & 0 & 0 \\ * & * & \Xi^a{}_{33} & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & h^2R - 2\iota P \end{bmatrix} < 0,$$

$\Pi^a{}_{11} = PA + A^{\mathrm{T}}P + \alpha P + Q - e^{-\alpha h}\mathcal{R}^a{}_{11}$, $\Pi^a{}_{12} = e^{-\alpha h}(\mathcal{R}^a{}_{11} - \mathcal{R}^a{}_{21}) - \lambda_i PBK$, $\Pi^a{}_{13} = e^{-\alpha h}\mathcal{R}^a{}_{21}$, $\Pi^a{}_{22}(\lambda_i) = e^{-\alpha h}(-\mathcal{R}^a{}_{11} + \mathcal{R}^a{}_{21} + (\mathcal{R}^a{}_{21})^{\mathrm{T}} - \mathcal{R}^a{}_{22}) + \theta\lambda_i\Omega + \lambda_i Z$, $\Pi^a{}_{23} = e^{-\alpha h}(-\mathcal{R}^a{}_{21} + \mathcal{R}^a{}_{22})$, $\Pi^a{}_{24}(\lambda_i) = -\theta\lambda_i\Omega - \lambda_i Z$, $\Pi^a{}_{33} = -e^{-\alpha h}(Q + \mathcal{R}^a{}_{22})$, $\Pi_{44}(\lambda_i) = \theta\lambda_i\Omega + \lambda_i Z - \frac{1}{\lambda_i}\Omega$, $\Xi^a{}_{11} = PA + A^{\mathrm{T}}P + Q - W - e^{-\alpha h}\mathcal{R}^a{}_{11}$, $\Xi^a{}_{12} = e^{-\alpha h}(\mathcal{R}^a{}_{11} - \mathcal{R}^a{}_{21})$, $\Xi^a{}_{22} = e^{-\alpha h}(-\mathcal{R}^a{}_{11} + \mathcal{R}^a{}_{21} + (\mathcal{R}^a{}_{21})^{\mathrm{T}} - \mathcal{R}^a{}_{22})$, $\Xi^a{}_{23} = e^{-\alpha h}(-\mathcal{R}^a{}_{21} + \mathcal{R}^a{}_{22})$, $\Xi^a{}_{33} = -e^{-\alpha h}(Q + \mathcal{R}^a{}_{22})$, $\mathcal{R}^1{}_{11} = 2R - \Upsilon_2{}^{\mathrm{T}}R^{-1}\Upsilon_2$, $\mathcal{R}^1{}_{21} = \Upsilon_1$, $\mathcal{R}^1{}_{22} = R$, $\mathcal{R}^0{}_{11} = R$, $\mathcal{R}^0{}_{21} = \Upsilon_2$, $\mathcal{R}^0{}_{22} = 2R - \Upsilon_1 R^{-1}\Upsilon_1{}^{\mathrm{T}}$, where $i = 1, 2, \ldots, N$, $a = 1, 0$, then the leader-following consensus can be achieved and the minimum upper bound of the communication delay satisfies $\bar{\eta} = \frac{\alpha h}{\alpha+\gamma}$, where $\gamma = [\lambda_{\max}(W)/\lambda_{\min}(P)]$.

Proof: Construct the following L-K function

$$V(t) = \varepsilon^{\mathrm{T}}(t)\bar{P}\varepsilon(t) + h\int_{-h}^{0}\int_{t+\tau}^{t} e^{\alpha(w-t)}\dot{\varepsilon}^{\mathrm{T}}(w)\bar{R}\dot{\varepsilon}(w)dwd\tau + \int_{t-h}^{t} e^{\alpha(w-t)}\varepsilon^{\mathrm{T}}(w)\bar{Q}\varepsilon(w)dw + \sum_{i=1}^{N}\delta_i(t),$$

with $\bar{P} = \mathcal{N}\otimes P$, $\bar{Q} = \mathcal{N}\otimes Q$, $\bar{R} = \mathcal{N}\otimes R$. We have $V(t) > 0$,

$$\begin{aligned}\dot{V}(t) \le{}& 2\varepsilon^{\mathrm{T}}(t)\bar{P}\dot{\varepsilon}(t) - \alpha h\int_{-h}^{0}\int_{t+\tau}^{t} e^{\alpha(w-t)}\dot{\varepsilon}^{\mathrm{T}}(w)\bar{R}\dot{\varepsilon}(w)dwd\tau \\ &+ h^2\dot{\varepsilon}^{\mathrm{T}}(t)\bar{R}\dot{\varepsilon}(t) - h\int_{t-h}^{t} e^{\alpha(w-t)}\dot{\varepsilon}^{\mathrm{T}}(w)\bar{R}\dot{\varepsilon}(w)dw \\ &+ \varepsilon^{\mathrm{T}}(t)\bar{Q}\varepsilon(t) - \alpha\int_{t-h}^{t} e^{\alpha(s-t)}\varepsilon^{\mathrm{T}}(w)\bar{Q}\varepsilon(w)dw \\ &- e^{-\alpha h}\varepsilon^{\mathrm{T}}(t-h)\bar{Q}\varepsilon(t-h) + \sum_{i=1}^{N}[-\delta_i(t) - \mu_i\delta_i(sm)] \\ &+ (e(mh) + \varepsilon(t-\eta(t)))^{\mathrm{T}}(\mathcal{N}^2\otimes Z)(e(mh) + \varepsilon(t-\eta(t))).\end{aligned} \tag{6}$$

Applying convex approach with (6), we can get the inequality:

$$-h\int_{t-h}^{t} e^{\alpha(w-t)}\dot{\varepsilon}^{\mathrm{T}}(w)\bar{R}\dot{\varepsilon}(w)dw \le -e^{\alpha h}\phi^{\mathrm{T}}(t)\begin{bmatrix}\bar{\mathcal{R}}_{11} & \bar{\mathcal{R}}_{12} \\ * & \bar{\mathcal{R}}_{22}\end{bmatrix}\phi(t),$$

where $\phi(t) = col\{\varepsilon(t) - \varepsilon(t-\eta(t)), \varepsilon(t-\eta(t)) - \varepsilon(t)\}$, $\bar{\mathcal{R}}_{11} = \mathcal{N}\otimes\mathcal{R}_{11}$, $\bar{\mathcal{R}}_{12} = \mathcal{N}\otimes\mathcal{R}_{12}$, $\bar{\mathcal{R}}_{22} = \mathcal{N}\otimes\mathcal{R}_{22}$, $\nu\in(0,1)$ and

$$\mathcal{R}_{11} = (1+\nu)R - \nu\Upsilon_2{}^{\mathrm{T}}R^{-1}\Upsilon_2, \quad \mathcal{R}_{21} = \nu\Upsilon_1 + (1-\nu)\Upsilon_2,$$
$$\mathcal{R}_{22} = (\nu-1)\Upsilon_1 R^{-1}\Upsilon_1 + (2-\nu)R.$$

1) When $\eta(t)\in[\eta, h)$, it is easy to get

$$\begin{aligned}&2\beta\dot{\varepsilon}^{\mathrm{T}}(t)\bar{P}[-(\mathcal{N}\otimes BK)\varepsilon(t-\eta(t)) \\ &\quad + (\mathcal{N}\otimes BK)e(mh) + (I\otimes A)\varepsilon(t) - \dot{\varepsilon}(t)] = 0.\end{aligned} \tag{7}$$

Then, if the DEM is not satisfied, it follows that

$$\begin{aligned}&\theta(\varepsilon(t-\eta(t)) - e(mh))^T(\mathcal{N}^2\otimes\Omega)(\varepsilon(t-\eta(t)) - e(mh)) \\ &\quad + \sum_{i=1}^{N}\mu_i\delta_i(t) - e^{\mathrm{T}}(mh)(I\otimes\Omega)e(mh) \ge 0.\end{aligned} \tag{8}$$

Define $\zeta(t) = col[\varepsilon(t), \varepsilon(t-\eta(t)), \varepsilon(t-h), e(mh), \dot{\varepsilon}(t)]$, combining (7), (8) with (6), we have

$$\dot{V}(t) \le -\alpha V(t) + \zeta^{\mathrm{T}}(t)\Phi^a\zeta(t) - \sum_{i=1}^{N}(1-\mu_i)\delta_i(t), \tag{9}$$

where

$$\Phi^a = \begin{bmatrix} \Phi^a{}_{11} & \Phi^a{}_{12} & \mathcal{N}\otimes e^{-\alpha h}\mathcal{R}^a{}_{21} & \mathcal{N}^2\otimes PBK & \beta\mathcal{N}\otimes PA \\ * & \Phi^a{}_{22} & \Phi^a{}_{23} & \Phi^a{}_{24} & \Phi^a{}_{25} \\ * & * & \Phi^a{}_{33} & 0 & 0 \\ * & * & * & \Phi_{44} & \Phi_{45} \\ * & * & * & * & \Phi_{55} \end{bmatrix} < 0,$$

$\Phi^a{}_{11} = \mathcal{N}\otimes(PA + A^{\mathrm{T}}P + \alpha P + Q - e^{-\alpha h}\mathcal{R}^a{}_{11})$, $\Phi^a{}_{12} = \mathcal{N}\otimes e^{-\alpha h}(\mathcal{R}^a{}_{11} - \mathcal{R}^a{}_{21}) - \mathcal{N}^2\otimes PBK$, $\Phi^a{}_{22} = \mathcal{N}\otimes e^{-\alpha h}(-\mathcal{R}^a{}_{11} + \mathcal{R}^a{}_{21} + (\mathcal{R}^a{}_{21})^{\mathrm{T}} - \mathcal{R}^a{}_{22}) + \mathcal{N}^2\otimes(\theta\Omega + Z)$, $\Phi^a{}_{23} = \mathcal{N}\otimes e^{-\alpha h}(-\mathcal{R}^a{}_{21} + \mathcal{R}^a{}_{22})$, $\Phi^a{}_{24} = -\mathcal{N}^2\otimes(\theta\Omega + Z)$, $\Phi^a{}_{25} = -\beta\mathcal{N}^2\otimes PBK$, $\Phi^a{}_{33} = -\mathcal{N}\otimes e^{-\alpha h}(Q + \mathcal{R}^a{}_{22})$, $\Phi_{44} = \mathcal{N}^2\otimes(\theta\Omega + Z) - (I\otimes\Omega)$, $\Phi_{45} = \beta\mathcal{N}^2\otimes PBK$, $\Phi_{55} = \mathcal{N}^2\otimes(h^2R - 2\beta P)$. Define $E = [E_1, E_2, \ldots, E_N]$, where $E_p = [e_p, e_{N+p}, \ldots, e_{N^2-N+P}]$, $p = 1, 2, \ldots, N$, e_i is a N-order matrix which the element of ith is 1 and the rest are 0. Let $\xi(t) = E^T diag\{U^T, U^T, \ldots, U^T\}\zeta(t)$, then, from the convex combination method, $\zeta^{\mathrm{T}}(t)\Phi^a\zeta(t) = \xi^{\mathrm{T}}(t)diag\{\lambda_1\Sigma^a(\lambda_1), \lambda_2\Sigma^a(\lambda_2), \ldots, \lambda_N\Sigma^a(\lambda_N)\}\xi(t) < 0$. Hence combining with (9), we have

$$\dot{V}(t) \le -\alpha V(t). \tag{10}$$

2) When $\eta(t)\in[0,\eta)$, the zero equation is $2\iota\dot{\varepsilon}^{\mathrm{T}}(t)\bar{P}[(I\otimes A)\varepsilon(t) - \dot{\varepsilon}(t)] = 0$. Define $\bar{W} = \mathcal{N}\otimes W$, substituting it into (6), we can get

$$\dot{V}(t) \le \gamma V(t) + \zeta^{\mathrm{T}}(t)\Psi^a\zeta(t), \tag{11}$$

where

$$\Psi^a = \begin{bmatrix} \Psi^a{}_{11} & \Psi^a{}_{12} & \mathcal{N}\otimes e^{-\alpha h}\mathcal{R}^a{}_{13} & 0 & \beta\mathcal{N}\otimes PA \\ * & \Psi^a{}_{22} & \Psi^a{}_{23} & 0 & 0 \\ * & * & \Psi^a{}_{33} & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & \Psi_{55} \end{bmatrix} < 0,$$

$\Psi^a{}_{11} = \mathcal{N}\otimes(PA + A^{\mathrm{T}}P + Q - W - e^{-\alpha h}\mathcal{R}^a{}_{11})$, $\Psi^a{}_{12} = \mathcal{N}\otimes e^{-\alpha h}(\mathcal{R}^a{}_{11} - \mathcal{R}^a{}_{21})$, $\Psi^a{}_{22} = \mathcal{N}\otimes e^{-\alpha h}(-\mathcal{R}^a{}_{11} + \mathcal{R}^a{}_{21} + (\mathcal{R}^a{}_{21})^{\mathrm{T}} - \mathcal{R}^a{}_{22})$, $\Psi^a{}_{23} = \mathcal{N}\otimes e^{-\alpha h}(-\mathcal{R}^a{}_{21} + \mathcal{R}^a{}_{22})$, $\Psi^a{}_{33} = -\mathcal{N}\otimes e^{-\alpha h}(Q + \mathcal{R}^a{}_{22})$, $\Psi_{55} = \mathcal{N}\otimes(h^2R - 2\iota P)$.

Similar with 1), we can get

$$\dot{V}(t) \le \gamma V(t). \tag{12}$$

Hence, from Lemma 2 and equations (10), (12), we can obtain $V(t) \le e^{-\alpha|\Theta_H(0,t)|}e^{\gamma|\Theta_D(0,t)|}V(0) \le e^{-\alpha t}e^{(\alpha+\gamma)(\frac{t}{h}+1)\eta}V(0) = \sigma e^{-\bar{\alpha}t}V(0)$, where $\sigma = e^{(\alpha+\gamma)\eta}$, $\bar{\alpha} = \alpha - \frac{\eta}{h}(\alpha+\gamma) > 0$, we can obtain $\eta < \frac{\alpha h}{\alpha+\gamma}$ (which can be guaranteed by $\bar{\eta} = \frac{\alpha h}{\alpha+\gamma}$), besides, $\gamma = [\lambda_{\max}(W)/\lambda_{\min}(P)]$, then, $\lambda_1\lambda_{\min}(P)||\varepsilon(t)||^2 \le V(t) \le \tau e^{-\bar{\alpha}t}V(0) \le \kappa\tau e^{-\bar{\alpha}t}||\varepsilon(0)||^2$. Hence $||\varepsilon(t)|| \le \frac{1}{\lambda_1}\kappa e^{-\frac{\bar{\alpha}}{2}t}||\varepsilon(0)||$, where $\kappa = \sqrt{(\tau(\lambda_{max}(\bar{P}) + h\lambda_{max}(\bar{Q})) + h^3\lambda_{max}(\bar{R})))/\lambda_{\min}(P)}$. It implies the system is exponential stable. Besides, changing the range of the initial state can't affect the convergence rate of the controller and the system judges whether the sampling signal is triggered after period h, that is $t^i_{k+1}h - t^i_k h \ge h$, there is no Zeno behavior. ■

Remark 2: Compared with [20], [21], we consider more complex time-varying communication delay. The current sampling signal $x_i(mh)$, the last trigger signal $x_i(t^i_k h)$ and the last trigger signal of the neighbor agent $x_j(t^j_{k'}h)$ are required when judging whether the DEM is satisfied. But $x_j(t^j_{k'}h)$ can't reach agent j at $t^j_{k'}h$ and the delay between adjacent agents will be different due to different channels. Therefore, the state signal $x_i(t^i_k h)$ is obtained by agent j at $t^i_k h + \eta^{ij}_k + \bar{\eta}$. The delay will

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cause asynchronous phenomenon of signal transmission and the switching controller can eliminate it.

Define $\tilde{P} = XPX$, $\tilde{Q} = XQX$, $\tilde{R} = XRX$, $\tilde{W} = XWX$, $\tilde{\Omega} = X\Omega X$, $\tilde{Z} = XZX$, $Y = KX$. The following result is presented.

Theorem 2: For positive scalers α, β, ι, θ and h, if there are matrices $\tilde{P} > 0$, $\tilde{Q} > 0$, $\tilde{R} > 0$, $\tilde{W} > 0$, $\tilde{\Omega} > 0$, $\tilde{Z} > 0$, $\tilde{\Upsilon}_1$, $\tilde{\Upsilon}_2$ and μ_i satisfies *Lemma 2* such that the following LMIs are satisfied:

$$\tilde{\Pi}^a(\lambda_i) = \begin{bmatrix} \tilde{\Pi}^a_{11} & \tilde{\Pi}^a_{12}(\lambda_i) & \tilde{\Pi}^a_{13} & \lambda_i BY & \tilde{\Pi}^a_{15} & \Gamma^a_1 \\ * & \tilde{\Pi}^a_{22}(\lambda_i) & \tilde{\Pi}^a_{23} & \tilde{\Pi}_{24}(\lambda_i) & \tilde{\Pi}^a_{25}(\lambda_i) & \Gamma^a_2 \\ * & * & \tilde{\Pi}^a_{33} & 0 & 0 & \Gamma^a_3 \\ * & * & * & \tilde{\Pi}_{44}(\lambda_i) & \beta\lambda_i BY & 0 \\ * & * & * & * & \tilde{\Pi}_{55} & 0 \\ * & * & * & * & * & -\tilde{R} \end{bmatrix} < 0,$$

$$\tilde{\Xi}^a = \begin{bmatrix} \tilde{\Xi}^a_{11} & e^{-\alpha h}(\tilde{\mathcal{R}}^a_{11} - \tilde{\mathcal{R}}^a_{21}) & e^{-\alpha h}\tilde{\mathcal{R}}^a_{21} & \iota AX & \Gamma^a_1 \\ * & \tilde{\Xi}^a_{22} & \tilde{\Xi}^a_{23} & 0 & \Gamma^a_2 \\ * & * & \tilde{\Xi}^a_{33} & 0 & \Gamma^a_3 \\ * & * & * & h^2R - 2\iota P & 0 \\ * & * & * & * & -\tilde{R} \end{bmatrix} < 0,$$

$\tilde{\Pi}^a_{11} = AX + XA^{\mathrm{T}} + \alpha X + \tilde{Q} - e^{-\alpha h}\tilde{\mathcal{R}}^a_{11}$, $\tilde{\Pi}^a_{12} = e^{-\alpha h}(\tilde{\mathcal{R}}^a_{11} - \tilde{\mathcal{R}}^a_{21}) - \lambda_i BY$, $\tilde{\Pi}^a_{13} = e^{-\alpha h}\tilde{\mathcal{R}}^a_{21}$, $\tilde{\Pi}^a_{15} = \beta AX$, $\tilde{\Pi}^a_{22}(\lambda_i) = e^{-\alpha h}(-\tilde{\mathcal{R}}^a_{11} + \tilde{\mathcal{R}}^a_{21} + (\tilde{\mathcal{R}}^a_{21})^{\mathrm{T}} - \tilde{\mathcal{R}}^a_{22}) + \theta\lambda_i\tilde{\Omega} + \lambda_i\tilde{Z}$, $\tilde{\Pi}^a_{23} = e^{-\alpha h}(-\tilde{\mathcal{R}}^a_{21} + \tilde{\mathcal{R}}^a_{22})$, $\tilde{\Pi}^a_{24}(\lambda_i) = -\theta\lambda_i\tilde{\Omega} - \lambda_i\tilde{Z}$, $\tilde{\Pi}^a_{25}(\lambda_i) = -\beta\lambda_i BY$, $\tilde{\Pi}^a_{33} = -e^{-\alpha h}(\tilde{Q} + \tilde{\mathcal{R}}^a_{22})$, $\tilde{\Pi}_{44}(\lambda_i) = \theta\lambda_i\tilde{\Omega} + \lambda_i\tilde{Z} - \frac{1}{\lambda_i}\tilde{\Omega}$, $\tilde{\Pi}_{55} = h^2\tilde{R} - 2\beta X$, $\tilde{\Xi}^a_{11} = AX + XA^{\mathrm{T}} + \tilde{Q} - \tilde{W} - e^{-\alpha h}\tilde{\mathcal{R}}^a_{11}$, $\tilde{\Xi}^a_{22} = e^{-\alpha h}(-\tilde{\mathcal{R}}^a_{11} + \tilde{\mathcal{R}}^a_{21} + (\tilde{\mathcal{R}}^a_{21})^{\mathrm{T}} - \tilde{\mathcal{R}}^a_{22})$, $\tilde{\Xi}^a_{23} = e^{-\alpha h}(-\tilde{\mathcal{R}}^a_{21} + \tilde{\mathcal{R}}^a_{22})$, $\tilde{\Xi}^a_{33} = -e^{-\alpha h}(\tilde{Q} + \tilde{\mathcal{R}}^a_{22})$, $\tilde{\mathcal{R}}^1_{11} = 2\tilde{R}$, $\tilde{\mathcal{R}}^1_{21} = \tilde{\Upsilon}_1$, $\tilde{\mathcal{R}}^1_{22} = \tilde{R}$, $\tilde{\mathcal{R}}^0_{11} = \tilde{R}$, $\tilde{\mathcal{R}}^0_{21} = \tilde{\Upsilon}_2$, $\tilde{\mathcal{R}}^0_{22} = 2\tilde{R}$, $\Gamma^1_1 = \tilde{\Upsilon}_2$, $\Gamma^1_1 = -\tilde{\Upsilon}_2$, $\Gamma^1_3 = 0$, $\Gamma^0_1 = 0$, $\Gamma^0_1 = \tilde{\Upsilon}_1$, $\Gamma^0_3 = -\tilde{\Upsilon}_1$, where $i = 1, N$, $a = 1, 0$, then the leader-following consensus can be achieved with a controller gain YX^{-1} and dynamic event-triggered parameters $\Omega = X^{-1}\tilde{\Omega}X^{-1}$.

Proof: According to $\lambda_i \leq \lambda_N$ and the Schur complement lemma, then the convex combination method is applied, we can verify that $\Pi^a(\lambda_i) < 0$ and $\Xi^a < 0$, which completes the proof. ■

V. Numerical Simulation

In this section, two simulations are done to illustrate the efficiency of the proposed dynamic event-triggered control scheme.

Simulation 1 (Numerical Simulation): The MASs in [23] is considered. It has one leader and four followers, where the topological structure is shown in Fig. 2(a) and

$$A = \begin{bmatrix} 0.003, 1, 0; & -1, 0.001, 0; & 0, 0, -2 \end{bmatrix};$$
$$B = \begin{bmatrix} 0.1 & 0.2 & 0.1 \end{bmatrix}^{\mathrm{T}}.$$

In this system, by manually adjusting the values of the parameters, the appropriate parameters are selected to make the system stable on the premise of ensuring the feasible solution of LMI. Next, we choose $\alpha = 0.11$, $h = 0.06s$. Let $\theta = 0.02$, $\mu_1 = 0.05$, $\mu_2 = 0.03$, $\mu_3 = 0.1$, $\mu_4 = 0.1$, $\bar{\delta}_1 = \bar{\delta}_2 = 0.01$, $\bar{\delta}_3 = \bar{\delta}_4 = 0.02$. The initial condition is $x_0(0) = [0.3, 0.2, -0.1]^{\mathrm{T}}$, $x_1(0) = [0.1, 0.2, 0.3]^{\mathrm{T}}$, $x_2(0) = [0.4, 0.5, 0.2]^{\mathrm{T}}$, $x_3(0) = [-0.1, -0.1, -0.2]^{\mathrm{T}}$, $x_4(0) = [-0.3, -0.1, -0.2]^{\mathrm{T}}$. Under these parameters, $\bar{\eta} = 0.0001s$. Fig. 2(b) shows the trajectory of each agent.

Fig. 2(c) shows the triggering instants of DEM and it can be found from Table I that compared with [13], [23], [27], the average trigger interval is longer. Compared with the ETM in [13], the DEM in this brief extends the average trigger time interval of each agent by 10%, 23.2%, 41% and 20%, respectively. Compared with [23], the average trigger time interval of each agent is extended by 4.7%, 6.2%, 18.2% and 9.1%, respectively, and compared with [27], it is extended by 1.7%, 13%, 32% and 11.3%, respectively. Moreover, Fig. 2(d) shows that the dynamic variables are gradually decayed.

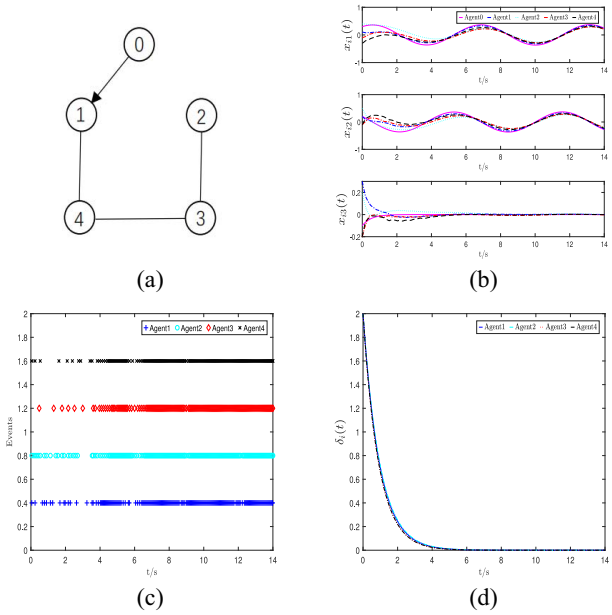


Fig. 2. (a) Topology graph $\mathcal{G}_1$ of MASs in Simulation 1. (b) Trajectories of the agents in Simulation 1. (c) Triggering instants of Simulation 1. (d) Change trend of $\delta(t)$ in Simulation 1.

TABLE I
Average Trigger Interval of Each Follower Agent

Method	Agent1	Agent2	Agent3	Agent4
[13]	0.0791	0.0745	0.0749	0.0773
[23]	0.0753	0.0642	0.0628	0.0700
[27]	0.0814	0.0700	0.0670	0.0757
This paper	0.0828	0.0791	0.0886	0.0843

Simulation 2 (Actual System Simulation): Considering the multiple damped oscillators, whose topological structure is shown in Fig. 3(a). The dynamic of leader oscillator and each following oscillator is

$$\begin{cases} \dot{p}_0(t) = v_0(t), \\ \dot{v}_0(t) = -\omega^2 p_0(t) - \delta v_0(t), \end{cases}$$
$$\begin{cases} \dot{p}_i(t) = v_i(t), \\ \dot{v}_i(t) = -\omega^2 p_i(t) - \delta v_i(t) + u_i(t), \end{cases}$$

where ω is the frequency of the oscillator and δ is the damping coefficient of the oscillator. Let $x_i(t) = [p_i(t), v_i(t)]^{\mathrm{T}}$, by linearizing the dynamic of oscillators, then $A = \begin{bmatrix} 0, 1; & -\omega^2, -\delta \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \end{bmatrix}^{\mathrm{T}}$.

In this example, let $\omega = 1$, $\delta = 1$, $\alpha = 0.42$, $h = 0.06s$, $\mu_1 = 0.25$, $\mu_2 = 0.3$, $\mu_3 = 0.2$, $\mu_4 = 0.7$, $\mu_5 = 0.7$, $\bar{\delta}_1 = \bar{\delta}_3 = 0.01$, $\bar{\delta}_2 = \bar{\delta}_4 = \bar{\delta}_5 = 0.02$. The initial condition is $x_0(0) = [0.3, 0.2]^{\mathrm{T}}$, $x_1(0) = [-0.1, 0.1]^{\mathrm{T}}$, $x_2(0) = [0.2, 0.3]^{\mathrm{T}}$, $x_3(0) = [0.4, 0.5]^{\mathrm{T}}$, $x_4(0) = [0.2, -0.1]^{\mathrm{T}}$, $x_5(0) = [0.5, -0.2]^{\mathrm{T}}$.

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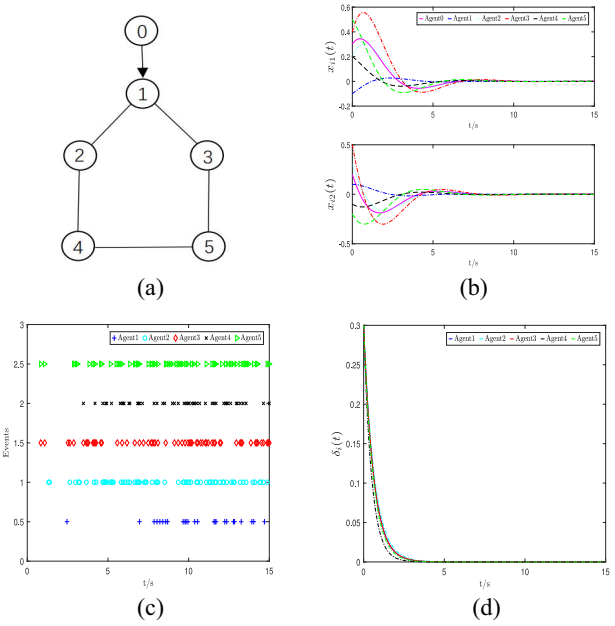


Fig. 3. (a) Topology graph $\mathcal{G}_2$ in Simulation 2. (b) Trajectories of the agents in Simulation 2. (c) Triggering instants of Simulation 2. (d) Change trend of $\delta(t)$ in Simulation 2.

TABLE II
AVERAGE TRIGGER INTERVAL OF EACH FOLLOWER AGENT

Method	Agent1	Agent2	Agent3	Agent4	Agent5
[23]	0.3191	0.1579	0.1515	0.1974	0.1500
This paper	0.6000	0.2206	0.2459	0.3125	0.2273

Under the parameters we can obtain $\bar{\eta} = 0.0178s$. Fig. 3(b) shows the trajectory of the damped oscillators. Fig. 3(c) gives the trigger moment of followers. From Table II we get the advantages of the DEM in this brief. Fig. 3(d) shows that $\delta(t)$ is monotonically decreasing.

VI. CONCLUSION

In this brief, the DEM makes the controller get fewer triggers while the system reaches stability faster. In addition, the more complex time-varying communication delay is considered, and the switching control is used to eliminate the delay. At last, the numerical simulation and actual system simulation are carried out. But the DEM in this brief doesn't reduce the number of triggers of the agent to the greatest extent and compare with the fixed topology, the MASs under the switching topology are more practical. Thus, in the future work, we will study the more efficient DEM of heterogeneous MASs under switching topology and discuss how to avoid collision in the process of agent travel.

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